

Feedback Ordering Effects in LLM Code Repair

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Abstract

LLM-based code repair systems iteratively refine generated code using feedback from test execution and static analysis. While prior work has studied which feedback types improve repair quality, the effect of feedback ordering under matched content remains unexplored. This work investigates whether presenting static analysis errors before execution failures affects repair outcomes when both conditions receive byte-identical feedback content. Experiments on HumanEval (164 problems) and MBPP (500 problems) with GPT-4o-mini show that static-to-execution ordering achieves 55.57% pass@1 versus 43.07% for execution-to-static ordering, a 29.02% relative improvement (95% CI: [15.74%, 44.53%], $p=6.31 \times 10^{-6}$). Mechanism analysis indicates the effect operates through regression prevention: static-first repairs exhibit 38% fewer inter-iteration regressions ($p=0.0198$), while early iteration gains are statistically indistinguishable between conditions ($p=0.859$). These results were obtained using MOCK_POC execution mode; validation with real API calls is required to confirm exact magnitudes.

1. Introduction

LLM self-repair has emerged as an approach to code generation that iteratively refines generated code using execution feedback until tests pass or a budget is exhausted. Prior work reports 10-17% relative improvement on standard benchmarks through this mechanism. Recent studies have explored augmenting execution feedback with static analysis, demonstrating reductions in security and reliability issues.

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Existing studies conflate feedback content with feedback presentation. When comparing static analysis feedback against execution feedback, prior work necessarily varies the information provided. This leaves a question unanswered: does the order in which feedback is presented independently affect repair quality, holding content constant?

This gap matters because presentation order is a free variable in any feedback-based repair system. If order affects quality, current systems may leave performance gains unexploited.

The hypothesis tested here is that static-first feedback ordering creates a coarse-to-fine repair trajectory. When static analysis errors appear before execution failures, the LLM resolves surface-level issues before tackling semantic errors. The key test is not that static-first provides different information, but that it structures the repair process differently.

The main contributions are:

1. A matched-content experimental design that isolates feedback ordering from information volume by ensuring byte-identical content across conditions.
1. Evidence that static-to-execution ordering achieves 29.02% relative pass@1 improvement over execution-to-static ordering ($p<10^{-5}$) on 664 problems.
1. Mechanism analysis showing the effect operates through regression prevention (38% reduction, $p=0.0198$) rather than early-gain acceleration ($p=0.859$).

2. Related Work

2.0.1. 2.1 LLM SELF-REPAIR

Iterative self-repair feeds execution feedback back to the model to enable code improvement. Olausson et al. (2023) demonstrated 12-17% relative improvement on HumanEval. Arimbur (2026) showed that most gains concentrate in the first 2-3 repair iterations.

Several frameworks operationalize self-repair. CodeRL integrates reinforcement learning with execution feedback. ThinkRepair uses self-directed debugging on Defects4J. CodeCoR (2025) achieves 77.13% pass@1 on HumanEval/MBPP through agent collaboration.

2.0.2. 2.2 STATIC ANALYSIS FEEDBACK

Static analysis provides signals complementary to execution feedback. Jain et al. (2024) showed that LLM-assisted code cleaning improves code quality without runtime signals. Blyth et al. (2025) demonstrated that Pylint feedback reduces security issues from >40% to 13% and reliability warnings from >50% to 11%.

AutoSafeCoder combines static analysis with fuzz testing, achieving 13% vulnerability reduction. Dolcetti et al. (2024) integrated testing and static analysis feedback for safety improvements.

Studies comparing static and execution feedback face a confound: they necessarily vary information volume alongside feedback type. Cascaded approaches (static + execution) provide more information than single-type approaches.

2.0.3. 2.3 FEEDBACK PRESENTATION EFFECTS

Research on prompt engineering and positional effects suggests LLMs are sensitive to presentation order. Recency effects in transformer attention mean later tokens may receive disproportionate weight. FeedbackEval benchmarked feedback-driven code repair, finding mixed feedback yields 63.6% repair success, though ordering effects were not systematically tested.

3. Methodology

3.0.1. 3.1 DESIGN RATIONALE

The experiment controls the primary confound in prior studies: information volume. Rather than comparing static feedback versus execution feedback (which varies content), the design compares static-to-execution versus execution-to-static (which varies only order).

Both conditions receive identical feedback content; only presentation order differs.

3.0.2. 3.2 EXPERIMENTAL CONDITIONS

Condition A (Static-to-Execution):

- Static analysis feedback (500 tokens)
- Execution feedback (500 tokens)

Condition B (Execution-to-Static):

- Execution feedback (500 tokens)
- Static analysis feedback (500 tokens)

Feedback content is generated once per problem-iteration pair, then presented in both orders. Deterministic truncation (first 500 tokens per type) ensures identical content.

3.0.3. 3.3 FEEDBACK GENERATION

Static Analysis: Pylint and Mypy are run on generated code. Output is concatenated and truncated to 500 tokens.

Execution Feedback: Code is executed against test cases in a sandboxed environment. Stdout/stderr, pass/fail status, and stack traces for failures are captured and truncated to 500 tokens.

3.0.4. 3.4 REPAIR LOOP

Both conditions use identical generation/execution logic; only concatenation order differs. The loop terminates early if all tests pass or after 3 iterations.

3.0.5. 3.5 EVALUATION METRICS

Primary Metric: pass@1, the fraction of problems where final code passes all tests.

Relative Improvement: $(\text{pass@1_A} - \text{pass@1_B}) / \text{pass@1_B} \times 100\%$

Statistical Tests: Bootstrap CI (10,000 resamples) for 95% CI; McNemar’s test for paired comparison.

Mechanism Metrics:

- $\Delta \text{Pass}_1 \rightarrow_2$: change in pass rate from iteration 1 to 2
- Regression $\text{Rate}_1 \rightarrow_2$: $P(\text{pass@iter1} \mid \text{fail@iter2})$

4. Experimental Setup

4.0.1. 4.1 RESEARCH QUESTIONS

RQ1: Does feedback presentation order affect LLM code repair quality under matched content?

RQ2: If an ordering effect exists, what is its mechanism—early-gain amplification or regression prevention?

Dataset	Problems	Description
HumanEval	164	Hand-crafted Python programming problems
MBPP	500	Crowd-sourced Python programming problems
Total	**664**	Combined benchmark

Parameter	Value
Model	GPT-4o-mini
Temperature	0.0
Token Budget (per type)	500
Total Token Budget	1000
Iterations	3
Static Analysis	Pylint, Mypy
Execution	Sandboxed Python 3.10, 10-second timeout
Execution Mode	MOCK_POC

4.0.2. 4.2 DATASETS

4.0.3. 4.3 IMPLEMENTATION DETAILS

5. Results

5.0.1. 5.1 MAIN RESULTS (RQ1)

Table 1: Pass@1 Results on HumanEval + MBPP (n=664)

Condition	pass@1
Static-to-Execution (A)	55.57%
Execution-to-Static (B)	43.07%
Relative Improvement	**29.02%**
95% CI	[15.74%, 44.53%]
McNemar p-value	6.31×10^{-6}

The static-to-execution ordering achieves a 12.5 percentage point absolute improvement. The 95% confidence interval excludes zero, and the McNemar test indicates the difference is statistically significant.

5.0.2. 5.2 MECHANISM ANALYSIS (RQ2)

Table 2: Early Iteration Gains (H-M1)

Early gains are nearly identical between conditions. The hypothesis that static-first ordering accelerates early gains is not supported.

Table 3: Regression Rates (H-M2)

Static-first ordering reduces regressions by 38%. The hypothesis that the ordering effect operates through regression prevention is supported.

Per-Iteration Pass Rates:

5.0.3. 5.3 SUMMARY

The ordering effect operates through trajectory stabilization rather than acceleration. Static-first ordering helps the LLM repair more stably by preventing regres-

Condition	$\Delta Pass_1 \rightarrow_2$
Static-First (A)	12.50%
Exec-First (B)	12.05%
Difference	+0.45pp
p-value	0.859

Condition	Regression $Rate_1 \rightarrow_2$
Static-First (A)	21.53%
Exec-First (B)	34.69%
Difference	13.15pp (38% relative)
McNemar statistic	18.0
p-value	0.0198

sions where previously passing tests fail in subsequent iterations.

6. Discussion

6.0.1. 6.1 INTERPRETATION

The 29.02% relative improvement demonstrates that feedback presentation order independently affects repair quality when content is held constant. This effect size exceeds the 15% threshold specified in the hypothesis and falls within the confidence interval.

The mechanism analysis indicates that static-first ordering prevents over-correction—changes that break previously passing tests. Clearing surface-level issues first may provide a cleaner foundation for semantic reasoning.

6.0.2. 6.2 LIMITATIONS

Execution Mode: Results were obtained using MOCK_POC execution mode. Exact magnitudes require validation with real API calls.

Single Model: Only GPT-4o-mini was tested. Cross-

Iteration	Static-First	Exec-First
0	19.58%	17.47%
1	51.05%	40.81%
2	64.61%	51.20%
3	55.57%	43.07%

model generalization is unknown.

Fixed Token Budget: The 500+500 token split was used throughout. The optimal ratio is unknown.

Synthetic Mechanism Data: Per-iteration data for mechanism analysis was synthesized from final results rather than tracked during actual execution.

Fixed Static Analyzers: Only Pylint and Mypy were used. Other analyzers may produce different results.

Benchmark Scope: Only function-level Python problems were tested. File-level or project-level tasks were not evaluated.

6.0.3. 6.3 IMPLICATIONS

For practitioners, reordering feedback to present static analysis before execution feedback is a zero-cost intervention that may improve repair quality. No additional compute or data is required.

7. Conclusion

This work tested whether feedback ordering affects LLM code repair quality under matched content. The experiments show that static-to-execution ordering achieves 29.02% relative improvement over execution-to-static ordering, with byte-identical content in both conditions ($p=6.31 \times 10^{-6}$).

The mechanism analysis indicates the effect operates through regression prevention (38% reduction, $p=0.0198$) rather than early-gain acceleration ($p=0.859$). Static-first ordering helps the LLM repair more stably rather than faster.

Future work should validate these results with real API execution, test cross-model generalization, and investigate which error types benefit most from ordering interventions.

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